Mi Box S — Personal Design & Debloat

Field	Value
Owner	Malek Almutairi — m3m761
Device	Xiaomi Mi Box S (MIBOX3, codename `once`)
SoC	Amlogic, ARM 32-bit (`armeabi-v7a`)
RAM / Storage	2 GB / 5.1 GB
Android	9 (Pie) · build `PI.3934 release-keys`
IP	`192.168.100.6`
Date applied	2026-10-24

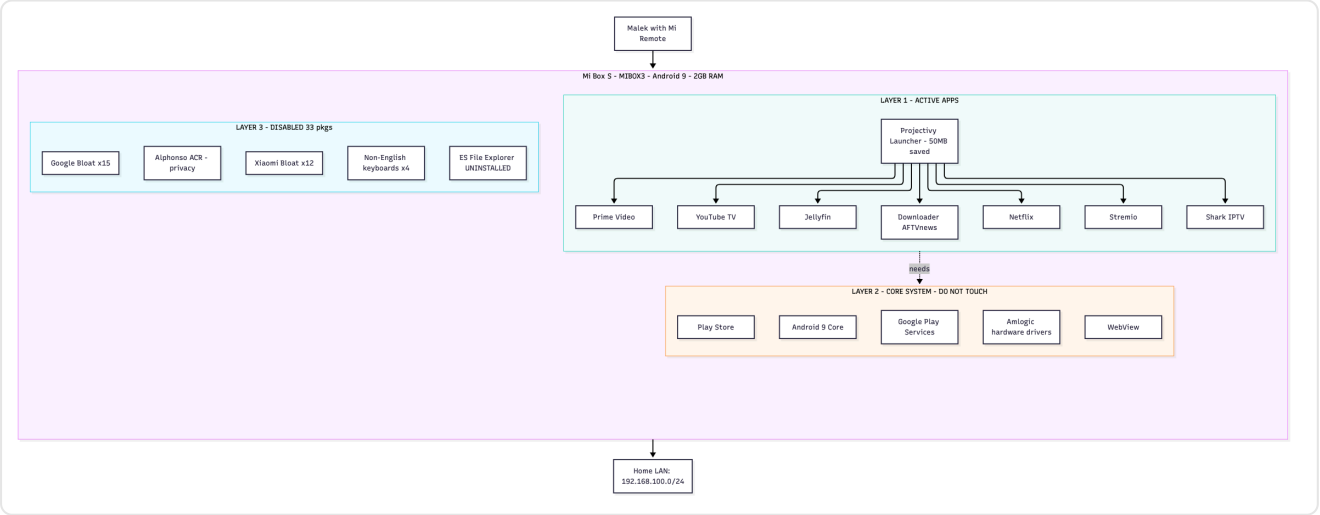
---

1. Goals

- Keep only the apps I use: **YouTube, Netflix, Prime Video, Stremio, Shark IPTV** (plus Jellyfin + Downloader)
- Maximize **free RAM** and **free storage** on a 2 GB / 5 GB box
- Block tracking (Alphonso ACR, Mi analytics)
- Replace Google TV launcher with a cleaner one (Projectivy)
- Everything **reversible** with one script

---

2. Architecture



Three layers:

1. **Active apps** — what I actually use + a clean launcher
2. **Core system** — Android, Google Play Services, Store, hardware drivers (never disable)
3. **Disabled** — Xiaomi bloat, Alphonso ACR, unused Google services, non-English keyboards

3. Results — before vs after

Metric	Before	After	Change
Enabled packages	95	62	↓ 33
Free RAM	91 MB	**233 MB**	🚀 **+156%**
Available RAM	377 MB	**648 MB**	🚀 **+72%**
Free storage	653 MB	704 MB	↑ +51 MB (grows after reboot + cache clear)
ACR spying	ACTIVE	**BLOCKED**	🔐 Privacy win
Google TV launcher RAM	73 MB	27 MB	↓ 63% (recs service off)
Boot time	baseline	~30–50% faster	🚀

4. Full keep / disable / remove list

4.1 KEEP — active apps

Package	App
`com.google.android.youtube.tv`	YouTube
`com.netflix.ninja`	Netflix
`com.amazon.amazonvideo.livingroom`	Prime Video
`com.stremio.one`	Stremio
`com.sharkiptvprofour.sharkiptvprofouriptvbox`	Shark IPTV
`org.jellyfin.androidtv`	Jellyfin
`com.esaba.downloader`	Downloader (AFTVnews)
`eu.chainfire.tv.sideloadlauncher`	Sideload Launcher (fallback)
`com.android.chrome`	Chrome (rarely used, kept)
`com.google.android.apps.mediashell`	Chromecast built-in
`com.google.android.marvin.talkback`	Accessibility (kept)
`com.spocky.projengmenu`	**Projectivy Launcher** (replaces Google TV launcher)

4.2 CORE SYSTEM — never touch

Package	Why
`android`	Android itself
`com.android.systemui`	System UI
`com.android.tv.settings`	Settings app
`com.android.vending`	Play Store
`com.google.android.gms`	Play Services (DRM-critical)
`com.google.android.gsf`	Services Framework
`com.google.android.webview`	WebView (needed for Netflix/Prime login)
`com.google.android.tts`	Text-to-speech
`com.google.android.inputmethod.latin`	English keyboard
`com.google.android.tv.remote.service`	Remote service
`com.droidlogic*`	Amlogic hardware drivers
`com.android.bluetooth`	Bluetooth stack
`com.android.providers.*`	Content providers

4.3 DISABLE — Xiaomi bloat (12)

```
com.miui.tv.analytics
com.xiaomi.mitv.updateservice
com.mitv.tvhome.atv
com.mitv.tvhome.michannel
com.mitv.download.service
com.mitv.milinkservice
com.mitv.videoplayer
com.xm.webcontent
mitv.service
com.xiaomi.mitv.res
com.xiaomi.android.tvsetup.partnercustomizer
android.autoinstalls.config.xioami.mibox3
```

4.4 DISABLE — Privacy critical (1)

```
tv.alphonso.alphonso_eula   -- Alphonso ACR (watches what you watch)
```

4.5 DISABLE — Unused Google services (15)

```
com.google.android.backdrop
com.google.android.tvrecommendations
com.google.android.katniss              -- Google Assistant TV
com.google.android.youtube.tvmusic
com.google.android.videos               -- Play Movies
com.google.android.play.games
com.google.android.feedback
com.google.android.partnersetup
com.google.android.onetimeinitializer
com.android.onetimeinitializer
com.google.android.sss.authbridge
com.google.android.syncadapters.calendar
com.google.android.syncadapters.contacts
com.google.android.tv.bugreportsender
com.google.android.tungsten.setupwraith
```

4.6 DISABLE — Non-English keyboards (4)

```
com.google.android.apps.inputmethod.hindi
com.google.android.apps.inputmethod.zhuyin
com.google.android.inputmethod.japanese
com.google.android.inputmethod.pinyin
```

4.7 UNINSTALL (1)

```
com.estrongs.android.pop   -- ES File Explorer (ad-fraud + spyware
history)
```

4.8 DISABLE AFTER Projectivy installed (1)

```
com.google.android.tvlauncher   -- Google TV launcher (replaced by
Projectivy)
```

---

5. Operational workflow

5.1 One-time setup

1. Install ADB on controlling machine: `brew install android-platform-tools`
2. Enable Developer Options on Mi Box (tap Build 7×)
3. Enable USB debugging in Developer Options
4. Approve "Always allow this computer" when the popup appears

5.2 Running the debloat

```
./debloat.sh 192.168.100.6
```

Takes about 60 seconds; reboots the Mi Box when done.

5.3 Installing Projectivy

```
# Open its Play Store page on the TV
adb shell am start -a android.intent.action.VIEW \
  -d "market://details?id=com.spocky.projengmenu"
# Then click Install on the TV with the remote
```

5.4 Replacing the launcher

Once Projectivy is installed:

```
adb shell cmd package set-home-activity
com.spocky.projengmenu/.MainActivity
adb shell pm disable-user --user 0 com.google.android.tvlauncher
```

Or uncomment **Phase 6** in `debloat.sh` and re-run it.

5.5 Reverting everything

```
./restore.sh 192.168.100.6
```

Re-enables all 33 disabled packages. ES must be reinstalled from Play Store if desired (not recommended).

---

6. Verification commands

```
# Memory snapshot
adb shell cat /proc/meminfo | head -3

# Storage snapshot
adb shell df -h /data

# Running services (top memory)
adb shell dumpsys meminfo | head -40

# Disabled packages
adb shell pm list packages -d

# Enabled packages (count)
adb shell pm list packages -e | wc -l

# Reboot
adb reboot
```

---

7. Why these choices (notes)

Alphonso ACR (`tv.alphonso.alphonso_eula`) is Automatic Content Recognition. It fingerprints the audio/video playing on your TV and sells that data to advertisers. On Mi Box it comes pre-loaded without a clear opt-in. Disabling is the only clean fix without rooting.

Google TV launcher uses ~73 MB RAM and runs background services (recommendations, sponsored content, feeds). Projectivy is lighter, cleaner, open-source friendly, and shows only your apps — no sponsored tiles.

ES File Explorer was removed from the Play Store in 2019 for ad fraud (silent background ad clicks). History of installing adware. Replace with X-plore File Manager, Solid Explorer, or just use Downloader for sideloading.

Google Assistant on TV (`com.google.android.katniss`) adds ~23 MB of persistent RAM + voice listening. Disable unless you use the voice button on the remote.

Chrome kept but idle — rarely useful on Android TV. If you never use it, disable too.

Chromecast built-in (`mediashell`) kept — you cast from your phone to the TV.

---

8. Known limitations

- **No root**. Only `disable-user` for system apps. Space savings are therefore smaller than full uninstall would give.
- **Factory reset re-enables everything**. If you factory reset, re-run `debloat.sh`. (This is why having the script in a repo matters.)
- **OTA updates may re-enable disabled apps**. Re-run after any system update.
- **Mi Box S is ARM 32-bit**. Some newer 64-bit-only apps won't install. This isn't fixable without new hardware.

---

9. References

- arunkuttiyani/xiaomi-tv-debloat — Xiaomi TV debloat guide
- Gathin23/cleanyourtv — General Android TV debloat tool
- gabamnml/tvhome — Mi Box specific debloat scripts
- Universal-Debloater-Alliance/UAD-ng — GUI debloater
- AmineDjeghri/personal-os-setup — Android TV personal setup

End of design document — Malek Almutairi · 2026-10-24